Revati Sanjay Chougule

bt24btech11018@iith.ac.in | revatichougule21@gmail.com | +91 8788324431 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.91
Shraddha Junior College of Science and Commerce,Ichalkaranji	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education ,kolhapur divisional board)	2024	83.67%
Navodaya Vidyalaya Padve,Rajapur,Ratnagiri,Maharashtra	Class X (Central Board of Secondary Education(CBSE))	2022	92.80%

PROJECTS

Customer Churn & Predictive Analytics Pipeline

Tech Stack: Python | Pandas | Scikit-Learn | Logistic Regression | SQL | SQLite | Matplotlib | Seaborn

- Developed an end-to-end **predictive analytics pipeline** using **Logistic Regression** to classify high-risk **customer churn** patterns.
- Executed **Exploratory Data Analysis (EDA)** and **SQL analytics** in **SQLite** for customer metrics, churn patterns, **missing-value handling**, and **behavioral analysis**.
- Evaluated classification performance using **Precision**, **Recall**, **F1-Score**, and **ROC-AUC**, and analyzed **Logistic Regression coefficients** through visual reports to derive customer churn insights.

API Rate Limiter & API Gateway

Tech Stack : Python | Object-Oriented Programming (OOP) | JSON | HTML | Matplotlib | Git | GitHub

- Developed a **Python-based API Rate Limiter & API Gateway** to simulate API traffic management using multiple rate-limiting algorithms.
- Implemented **Token Bucket, Fixed Window, Sliding Window, and Leaky Bucket** algorithms with configurable request limits through a JSON configuration file.
- Designed a modular system with **request logging, authentication support, performance statistics, and automated HTML/JSON report generation**.
- Generated **traffic analytics and visualization charts** using Matplotlib to monitor request distribution, success rate, and user statistics.

Data Engineering & Analytics Platform

Tech Stack: Python | Pandas | FastAPI | SQL | SQLite | React | ECharts | JavaScript

- Developed an **ETL pipeline** using Python and Pandas for data cleaning, validation, transformation, and processing.
- Developed **FastAPI REST APIs** with SQL analytics and SQLite for sales, category, region, and monthly analysis.
- Created an **interactive React + ECharts dashboard** with KPIs, filters, charts, and sales records.

AI Code Intelligence and Analysis Platform

Tech Stack: Python | AST | Google Gemini API | JSON | Matplotlib | HTML | Git | GitHub

- Developed an **AI-powered Python code analysis platform** combining AST-based static analysis with Google Gemini for code quality, maintainability, software metrics, and security analysis.
- Automated detection of **code smells and vulnerabilities**, including duplicate functions, dead code, unused variables/imports, long functions, missing docstrings, nested loops, and hardcoded passwords.
- Computed **LOC, Cyclomatic Complexity, function/class metrics, maintainability score, security score, and overall code health** for project-level analysis.

- Generated **HTML dashboards, JSON reports, visualization charts, and Gemini-powered code reviews** with summaries, finding explanations, security assessments, and improvement recommendations.

RELEVANT COURSES

Data Structures & Algorithms,Introduction to Programming,Probability and Statistics,Linear Algebra

Big Data Biology and Biological Databases,R and Python for Biologists, Next Generation Sequencing,Machine Learning for Bioinformatics

SKILLS

Programming Languages: Python,C,C++,SQL,R

Core Computer Science : Data Structures and Algorithms, Object-Oriented Programming(OOP), Database Management Systems(DBMS),Operating Systems(OS),Computer Networks(CN)

Machine Learning

Operating Systems**:**Linux (Ubuntu)

Data Science: Logistic Regression, EDA, Statistical Analysis, Model Evaluation (Precision, Recall, F1-Score, ROC-AUC)

Data Visualization: Matplotlib, Seaborn, ECharts, Dashboard Development

EXTRACURRICULAR

Represented IIT Hyderabad in Volleyball at the **Inter IIT Sports Meet (2025)**, demonstrating teamwork, discipline, and competitive sportsmanship.